

3-D magnetic equivalent circuit model for a coreless axial flux permanent-magnet synchronous generator

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Abstract

In coreless axial flux permanent magnet synchronous generators (AFPMSGs), magnetic leakage and fringing effect is significant due to the large air gap and increases the difficulty in magnetic field calculation. This paper presents a 3-D magnetic equivalent circuit (MEC) model for the magnetic field calculation in coreless AFPMSGs by considering the magnetic leakage and fringing flux. Field distribution is calculated based on the magnetic circuit analysis. The magnetic flux in the air gap between the permanent magnets is analysed using the magnetic field division method. Compared with the traditional magnetic circuit model, the magnetic leakage from PM to the rotor iron plates is considered in both radial and tangential directions in the proposed model. The proposed MEC method is applied to a specific AFPMSG, and the magnetic field distribution, no-load back electromotive force (EMF), and torque are calculated and compared with finite elements (FEM) results. AFPMSG machines with different pole-arc ratios are prototyped to validate the effectiveness and accuracy of the proposed 3-D MEC model. The results obtained by the proposed MEC model are in good agreement with those of the FEM and experimental results.

1 | INTRODUCTION

Axial flux permanent magnet (AFPM) machines with a single coreless stator and double rotors are widely used for wind power generators [1]. Compared to a radial flux electric machine, the coreless axial flux permanent magnet synchronous generator (AFPMSG) has the advantages of a larger power-to-weight ratio, high compactness, and better cooling [2]. The efficiency of the AFPMSG for wind power generation is high due to the absence of iron loss. The elimination of cogging torque leads to a small starting torque which will improve the efficiency of capturing wind energy [3–7].

The electromagnetic characteristics of a magnetic machine can be calculated based on the distribution of the magnetic field. The computation of the magnetic field is essential for the analysis and optimization in AFPMSG design. The most typical magnetic field analysis methods can be classified into the finite-element method (FEM), the analytical method, and the magnetic equivalent circuit (MEC) method.

The FEM is a precise modelling method for the numerical analysis of field distribution. 3-D FEM models have been proposed and used for the analysis of different types of AFPMSGs [8–11]. Moreover, the 3-D FEM model of a coreless AFPMSG is compared with a 2-D FEM model [12]. Although the FEM method has the advantage of high accuracy, it is often time-consuming and requires a very large amount of computing resources.

A number of analytical methods for magnetic field calculation in AFPMSGs have been reported by researchers [13–19]. An analytical mean of the radius model for a coreless single-side AFPM generator by calculating the scalar magnetic potentials using the Fourier series method is proposed in [14]. Ravaut [17] presented field distribution created by tile permanent magnets (PMs), which is a basic component in AFPM machines. Nasar and Xiong [18] proposed an analytical solution of the field distribution of a PM disk machine in the three spatial coordinates based on the concept of magnetic charges and the method of images. The air gap

magnetic fields of two generators are analysed by the analytical method for an AFPMSG with a Halbach permanent magnet array [19].

The MEC method is one convenient way to consider simultaneously as many effects as possible in magnetic field calculation for magnetic machines. It has a negligible computation time, but the accuracy strongly depends on the ratio of the cross-section and gap length. In AFPMSG machines, the air gap length is usually relatively large. The magnetic field distribution with a large air gap length is hard to be modelled by a 2-D model accurately. Therefore, the field distribution in the radial direction should be considered in magnetic field calculation for AFPM machines [20–22]. In iron-cored machines, since the stator coils are placed in the slots, the air-gap is treated as a constant and independent of the stator axial length. However, the air-gap length is dependent on the axial thickness of the stator for coreless AFPM machines, which enhances the complexity in the design process. Moreover, the coreless structure enhances the edge effect of the magnetic flux in AFPMSGs. It increases the difficulties in calculating the magnetic permeance of the magnetic leakage flux. An MEC model is established for the electromagnetic analysis of a coreless AFPMSG machine, but the amount of leakage flux and the calculation method is not specified obviously [20]. An improved MEC model is proposed for magnetic field calculation and evaluation of the air-gap leakage flux in AFPMSGs. The magnetic flux in the air gap is divided into different layers in the radial direction [23].

This paper proposes a 3-D MEC model of the AFPMSG by considering the magnetic leakage and fringing flux. In the proposed MEC model, the magnetic field distribution in 3 dimensions is calculated by detailed modelling of the magnetic leakage and magnetic flux in the air gap. The magnetic flux in the air gap between the permanent magnets is analysed using the magnetic field division method. The magnetic leakage from PM to the rotor iron plates is considered in both radial and tangential directions. The proposed MEC method is applied to a specific AFPMSG, and the magnetic field distribution, no-load back electromotive force (EMF), and torque are calculated and compared with the FEM results. AFPMSG machines with different pole-arc ratios are prototyped to validate the effectiveness and accuracy of the proposed 3-D MEC model.

2 | MAGNETIC EQUIVALENT CIRCUIT MODELLING AND ANALYSIS

2.1 | Machine Configurations

A schematic of a typical coreless AFPMSG is shown in Figure 1. The coreless AFPMSG consists of two external rotors and one internal coreless stator with three-phase windings. In each rotor, permanent magnets are mounted on the inner surface of a soft magnetic steel disk. The disk-shaped stator armature winding without an iron core is sandwiched between the magnets. Each armature winding has a trapezoidal

shape and is encapsulated in epoxy resin for mechanical strength.

In order to reduce the complexity of the computation, assumptions are made in our proposed 3-D MEC model:

- (1) The magnetic material has uniform magnetization and the relative permeability, μ_r , is considered a constant. The relative permeability of the PM is assumed to be 1 for the NdFeB sintered magnet.
- (2) Magnetic saturation is absent and the rotor iron plates have infinite magnetic permeability.

2.2 | 3-D magnetic equivalent circuit model

The magnetic flux distributions of the coreless AFPMSGs are shown in Figure 2. Only two neighbour PM pairs structures are shown, and a periodic boundary condition is used due to the symmetry. In the tangential section view, two kinds of magnetic leakage are considered in the model as shown in Figure 2 (a). The leakages between the PM and the iron rotor plate in the tangential direction are represented by P_{lc} . The leakages between the adjacent PMs are represented by P_{lp} . The main magnetic flux between the opposite PMs is represented by P_m . In the radial section view, the leakages between PMs and the iron rotor plate in the radial direction are represented by P_{lr} .

According to the distribution of magnetic flux, the MEC model is shown in Figure 3. The PM on the rotor is between the winding coil plate and the rotor iron plate. The main magnetic reluctances in the MEC model mainly consist of three parts: 1) the air gap; 2) the rotor; and 3) the stator. The magnetic flux paths in coreless AFPM machines can be divided into two categories: the main magnetic flux path and the leakage flux path. The main magnetic flux passes through the air gap between the opposite PMs on the rotor iron plates. The leakage flux passes through the neighbouring PMs or from the PM to the rotor iron plate.

The schematic view of the magnetic flux distribution from the tangential and radial directions of the coreless AFPMSG is shown in Figure 2 (a) and (b). The primary axial magnetic flux between the opposite PMs are derived into different parts, in which the magnetic permeances are represented by P_{g1} , P_{g2} , P_m , and so on. The magnetic leakage flux from PM to the rotor iron plate is considered separately in the tangential direction and the radial direction. The magnetic permeance for leakages in the tangential section is represented by P_{lc} , as shown in Figure 2 (a). The magnetic permeances for leakages in the radial section are represented by P_{lr} and P_{lb} , as shown in Figure 2 (b). P_{lp} refers to the magnetic leakage between the neighbouring PMs. The magnetomotive force (MMF) of the PMs is represented by F_m .

An overall 3-D MEC model is shown in Figure 3, and the coordinates are marked in the figure. Different parts of the air gap permeance are distinguished by a very thin dotted line. Air gap permeances are coloured in red and leakage permeances are coloured in yellow.

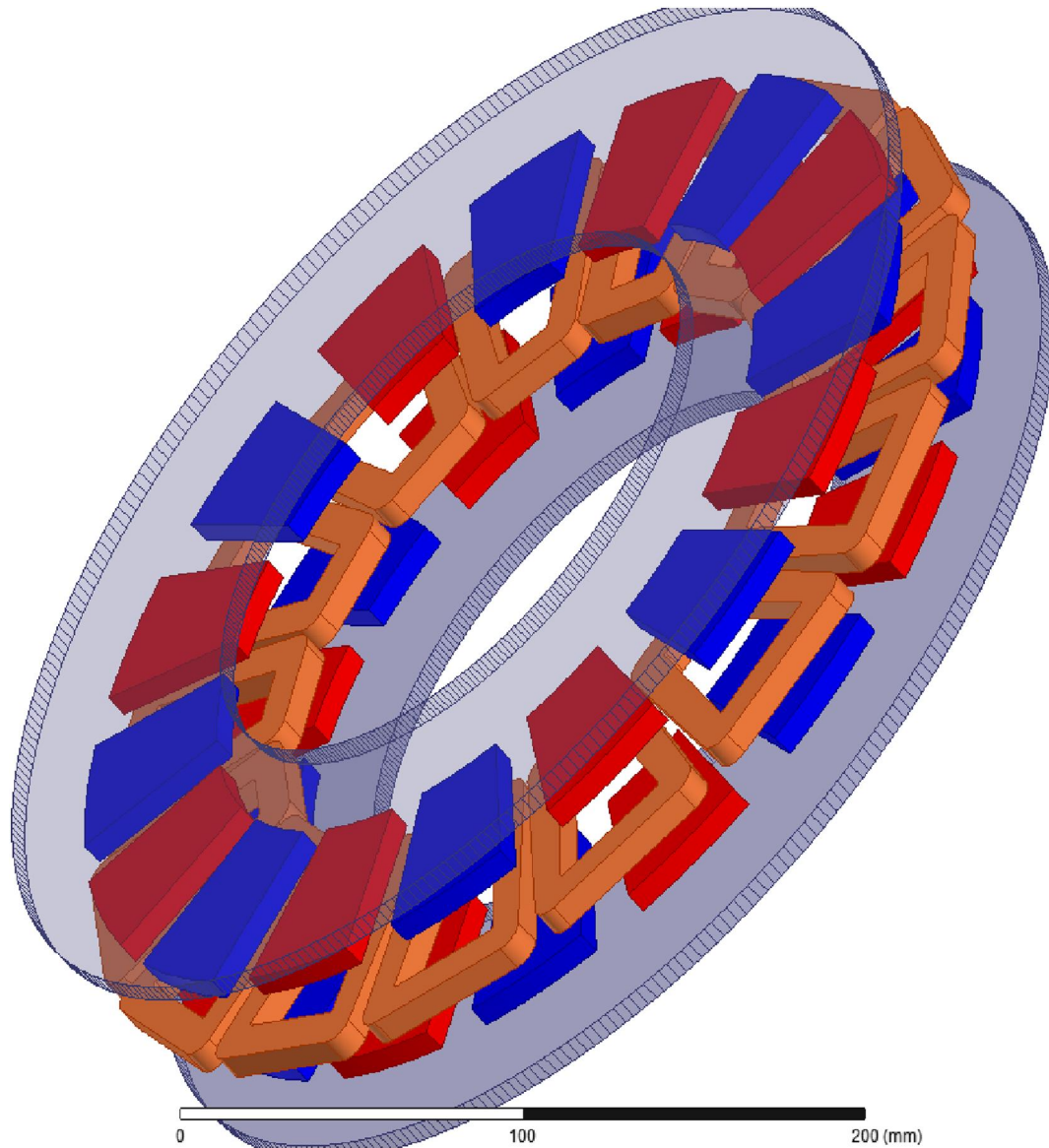


FIGURE 1 3-D schematic of a 16-poles coreless axial flux permanent magnet synchronous generators

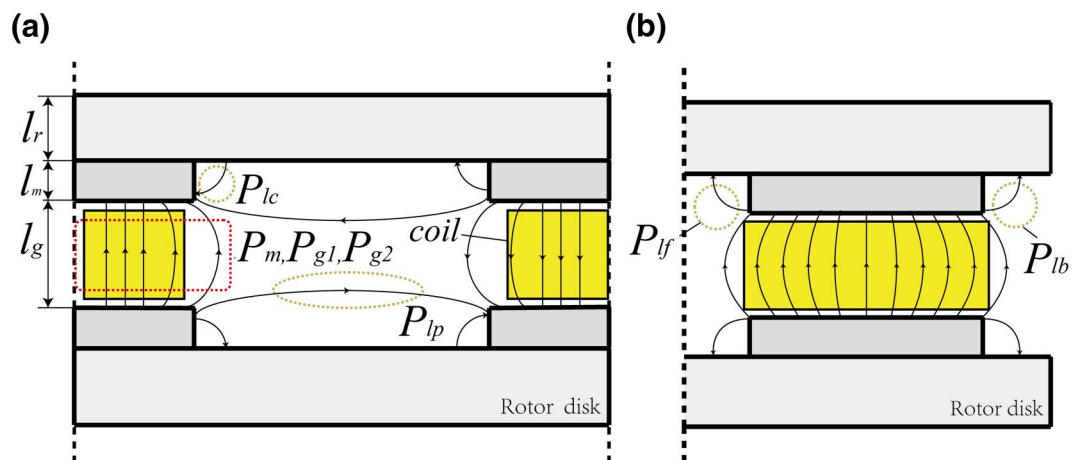


FIGURE 2 Magnetic flux distribution of a coreless axial flux permanent magnet synchronous generator. (a) tangential section view, (b) radial section view

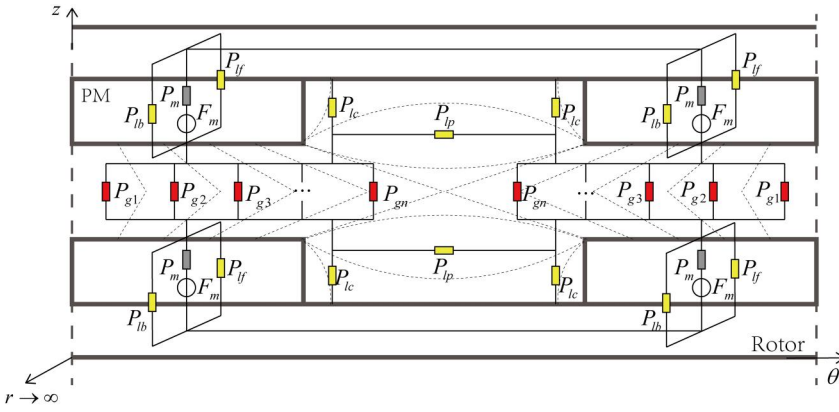


FIGURE 3 3-D magnetic equivalent circuit model for a coreless axial flux permanent magnet synchronous generators

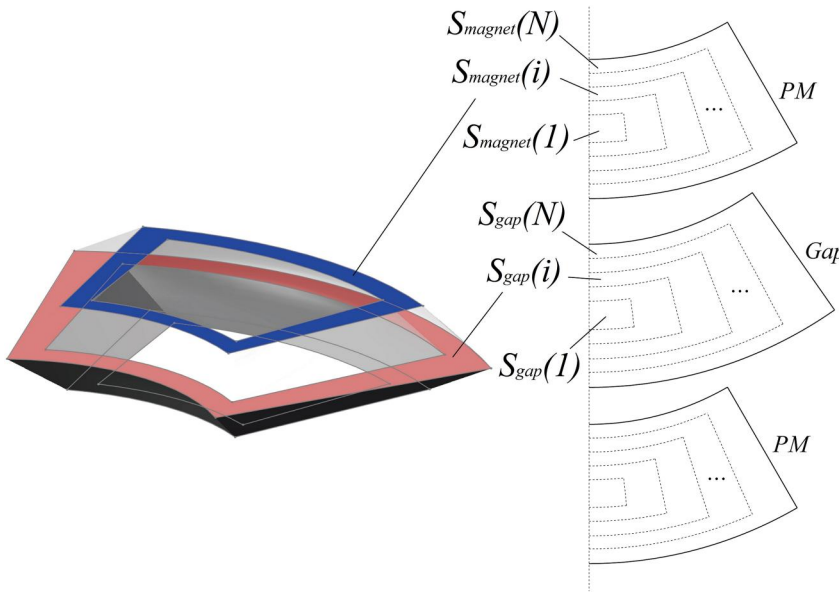


FIGURE 4 Schematic view of multi-level magnetic flux tubes in the air gap

2.3 | Calculation of the air-gap magnetic permeance

Since the magnetic flux passes through the interior of the air gap with a larger distance, the distribution of the magnetic field in the air gap should be considered separately to improve the accuracy of magnetic permeance calculation. An onion-shaped model is proposed to calculate the magnetic permeance of the air gap. The primary axial magnetic flux between the opposite PMs is derived into N parts, the choice of N depends on the accuracy of calculation; the magnetic permeances are modelled by $P_g(1), P_g(2), \dots, P_g(N)$, respectively. Each slice represents a part of the magnetic flux, and the shape of each part can be considered a geometry like the hollow frustum. To simplify the calculation of the volume of magnetic flux tubes, the lateral surfaces of the magnetic flux tube are assumed to be flat. The onion-shaped magnetic flux tubes are divided into N parts with different shapes, which are numbered from the inner side to the outer side from 1 to N . The 3-D schematic diagram of a single magnetic permeance tube and the cross-sectional areas S_{gap} and S_{magnet} of the interface are shown in Figure 4. The

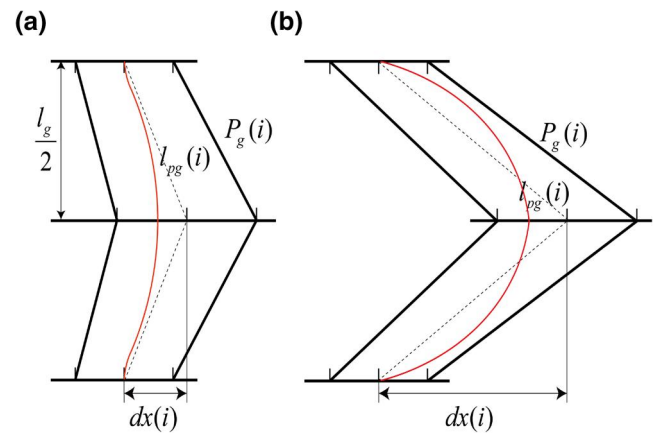


FIGURE 5 Side view of magnetic flux tubes. (a) $2dx(i) \leq l_g$, (b) $2dx(i) > l_g$

tangential cross-section of the i th magnetic flux tube is shown in Figure 5. The magnetic permeance of each magnetic flux tube can be written as:

$$P_g(i) = \frac{\mu_0 k_v V(i)}{l_{pg}(i)^2} \quad (1)$$

where $V(i)$ is the volume of the i_{th} magnetic flux tube. The volume correction factor, k_v , of the i_{th} magnetic flux tube in the air gap, which is assumed to be $k_v = 1.5$ in this work.

The volume of the magnetic flux tube can be calculated as:

$$V(i) = \frac{l_g (S_{magnet}(i) + S_{gap}(i) + \sqrt{S_{gap}(i) S_{magnet}(i)})}{3} \quad (2)$$

The area of the i_{th} magnetic flux tube at the PM surface and at the centre of the air gap are expressed, respectively:

$$S_{magnet}(i) = \frac{\pi \alpha_p (2i-1) (R_{mo}^2 - R_{mi}^2)}{4pN^2} \quad (3)$$

$$S_{gap}(i) = \frac{\pi (2i-1) (R_o^2 - R_i^2)}{4pN^2} \quad (4)$$

As shown in Figure 5, $dx(i)$ refers to the tangential distance between the centre of the air gap interface and the centre of the PM surface in the i_{th} magnetic flux tube. The value of $dx(i)$ can be expressed as:

$$dx(i) = \frac{\pi (1 - \alpha_p) \cdot (R_{mo} + R_{mi}) \cdot (i - 0.5)}{4pN} \quad (5)$$

where $l_{pg}(i)$ is the mean length of the i_{th} magnetic flux tube (represented by the red line in Figure 5). The expression of $l_{pg}(i)$ is divided into two distinct regions according to the relationship length of $dx(i)$ and half the gap length $\frac{l_g}{2}$. When $dx(i) \leq \frac{l_g}{2}$, the mean length of the magnetic flux tube is assumed to be equal to the length of the dotted arc line as shown in Figure 5 (a). Thus, $l_{pg}(i)$ can be written as:

$$l_{pg}(i) = \frac{l_g}{\sin \left[\arctan \left(\frac{2dx(i)}{l_g} \right) \right]} \arctan \left(\frac{2dx(i)}{l_g} \right) \quad (6)$$

When $dx(i) > \frac{l_g}{2}$, the expression of $l_{pg}(i)$ can be written as:

$$l_{pg}(i) = 2k_{p2}(i) \sqrt{dx(i)^2 + \frac{l_g^2}{4}} \quad (7)$$

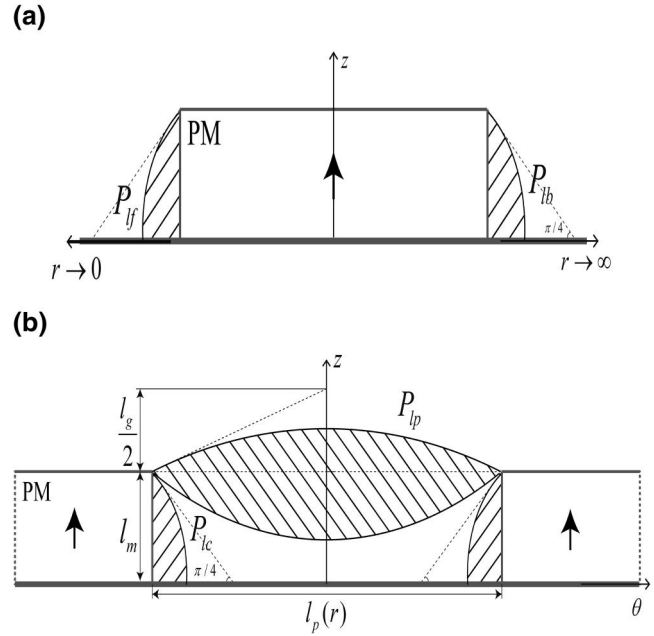


FIGURE 6 View of leakage flux permeance. (a) radial section view, (b) tangential section view

where $k_{p2}(i)$ is a linear correction coefficient to approximate the average length of the outer magnetic flux tube. The value of $k_{p2}(i)$ depends on the ratio of the linear distance between the centre of the dotted line, as shown in Figure 5 (b). The actual length of the i_{th} magnetic flux tube is expressed as:

$$k_{p2}(i) = \frac{0.7}{\frac{2dx(i)}{l_g} - 1} \left(\frac{2dx(i)}{l_g} - 1 \right) + 0.8 \quad (8)$$

2.4 | Calculation of magnetic leakage

Figure 6 shows the distribution of the magnetic leakage flux from the PM to the rotor. The leakage permeance P_{lp} between the neighbouring PMs can be calculated as:

$$P_{lp} = \frac{\mu_0 (R_{pout} - R_{pin}) \cdot (S_{pin} + S_{pout} + \sqrt{S_{pin} S_{pout}})}{3l_{p1}^2} \quad (9)$$

Leakage permeance from the PM to the rotor can be calculated as:

$$P_{lc} = \frac{\mu_0 (\pi - 2) \cdot (R_{pout} - R_{pin})}{6} \quad (10)$$

$$P_{lf} = \frac{\mu_0(\pi - 2) \cdot R_{mi} \sin\left(\frac{\pi\alpha_p}{2p}\right)}{6} \quad (11)$$

$$P_{lb} = \frac{\mu_0(\pi - 2) \cdot R_{mo} \sin\left(\frac{\pi\alpha_p}{2p}\right)}{6} \quad (12)$$

where $l_p(R_{mi})$, $l_p(R_{mo})$, θ_{in} , θ_{out} , S_{pin} , S_{pout} are the shape coefficients of the leakage flux permeance, $l_p(R_{mi})$ and $l_p(R_{mo})$ refer to the length of the air gap between the neighbouring poles of the inner edge and the outer edge, respectively. θ_{in} and θ_{out} are the central angle corresponding to the upper edge and the lower edge of P_{lc} . The values of the above parameters are calculated as,

$$l_p(R_{mi}) = 2R_{mi} \sin\left(\frac{\pi(1 - \alpha_p)}{2p}\right) \quad (13)$$

$$l_p(R_{mo}) = 2R_{mo} \sin\left(\frac{\pi(1 - \alpha_p)}{2p}\right) \quad (14)$$

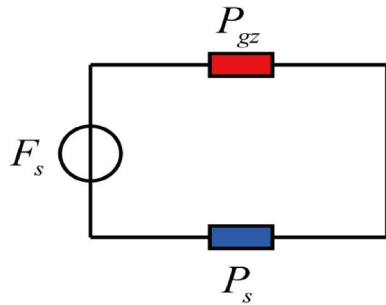


FIGURE 7 Simplified magnetic equivalent circuit model

$$l_{p1} = 0.75[l_p(R_{mi}) + l_p(R_{mo})] \quad (15)$$

$$\theta_{in} = \arctan\left(\frac{l_g}{l_p(R_{mi})}\right) \quad (16)$$

$$\theta_{out} = \arctan\left(\frac{l_g}{l_p(R_{mo})}\right) \quad (17)$$

$$R_{pin} = \frac{l_p(R_{mi}) \sin(\theta_{in})}{2} \quad (18)$$

TABLE 1 Parameters of the AFPMGS under study

Parameters	Symbol	Value
Inner PM radius	R_{mi}	210.8 mm
Outer PM radius	R_{mo}	314.8 mm
Rotor height	l_r	10 mm
Air-gap	l_g	10 mm
PM height	l_m	7 mm
Wire diameter	D_c	1 mm
Number of coils	N_c	15
Rated speed	ω_r	20.93c rad/s
Remanence	B_r	1.3 T
Coercivity	H_c	990 kA/m
Number of turns per phase	N_s	49
Phase resistance	R_a	10 Ω
Pole arc ratio	α_p	0.7
Pole-pairs	P	8

Abbreviations: AFPMGS, axial flux permanent magnet synchronous generators; PM, permanent magnet.

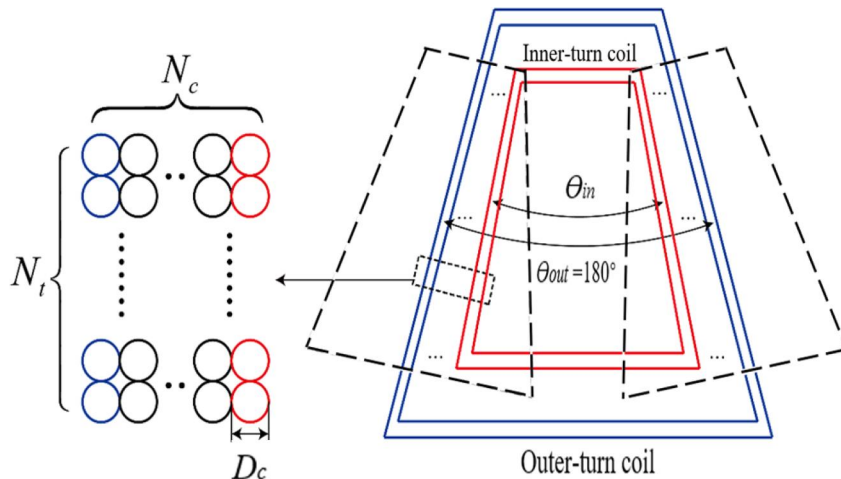


FIGURE 8 Intersection surface and the vertical view of a single coil

$$R_{pout} = \frac{l_p(R_{mo})\sin(\theta_{out})}{2} \quad (19)$$

$$S_{pin} = R_{pin}^2 \theta_{in} - \frac{l_p(R_{mi})[2R_{pin} \cos(\theta_{in}) + l_p(R_{mi})(2\pi - 2)]}{4} \quad (20)$$

$$S_{pout} = R_{pout}^2 \theta_{out} - \frac{l_p(R_{mo})[2R_{pout} \cos(\theta_{out}) + l_p(R_{mo})(2\pi - 2)]}{4} \quad (21)$$

The magnet permeance of the PM, P_m , is calculated as:

$$P_m = \frac{B_r \sum_{i=0}^N S_{magnet}(i)}{H_c l_m} \quad (22)$$

2.5 | Magnetic flux density distribution in the air gap

By using the Norton's and Thévenin's theorems, the MEC model in Figure 3 can be simplified as a series magnetic circuit shown in Figure 7.

The total primary air gap permeance P_{gz} is determined by the accumulation of segmented primary magnetic permeances as,

$$P_{gz} = \sum_{i=0}^N P_g(i) \quad (23)$$

P_s depends on the permeance of the PM and leakage permeance as,

$$P_s = \frac{P_m + P_{lb} + P_{lf} + P_{lc}}{2} + P_{l1} \quad (24)$$

Magnetomotive force F_s in the 3-D MEC model is given by,

$$F_s = \frac{H_c l_m P_m}{P_s} \quad (25)$$

The magnetic flux at the centre of the air-gap is considered to be perpendicular to the rotor disk. This part of the magnetic flux mainly contributes to the EMF. The value of the magnetic flux density at the centre of each element can be expressed as,

$$B_g(i) = \frac{F_s P_g(i)}{S_{gap}(i)(P_s + P_{gz})} \quad (26)$$

The main specification of the coreless AFPMMSG is given in Table 1. In the MEC model, the number of elements in the air gap permeance, N , is set to be 10. The calculated magnetic

TABLE 2 Calculated magnetic permeance for different magnetic flux tubes ($N = 10$)

Symbol	Value (μH)
$P_g(1)$	0.0022
$P_g(2)$	0.0066
$P_g(3)$	0.0108
$P_g(4)$	0.0145
$P_g(5)$	0.0179
$P_g(6)$	0.0209
$P_g(7)$	0.0222
$P_g(8)$	0.0133
$P_g(9)$	0.0087
$P_g(10)$	0.0060
P_p	0.0071
P_{lc}	0.0083
P_{lf}	0.0046
P_{lb}	0.0068
P_m	0.163

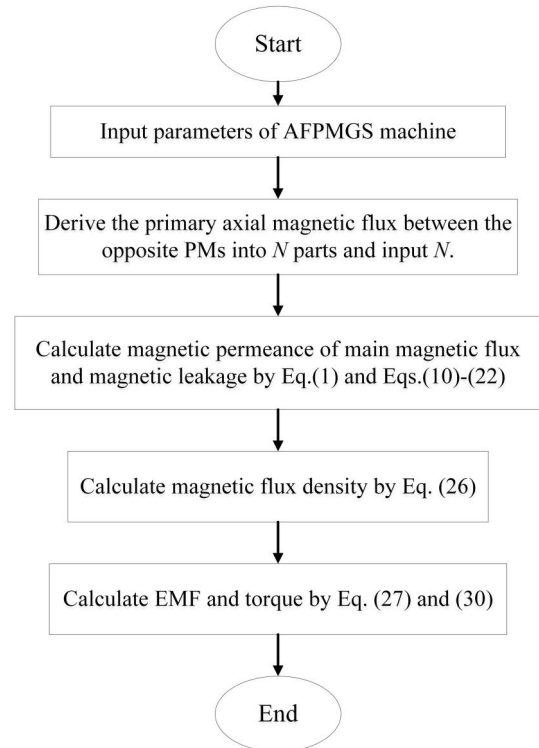


FIGURE 9 The flow chart of the proposed magnetic equivalent circuit method

permeance for different magnetic flux tubes by the MEC is listed in Table 2.

The correction factor is always used for magnetic field calculation of magnetic machines. But the correction factor seriously affects the accuracy of MEC models and its value is always empiric. The proposed MEC model can be used for the correction factor calculation in coreless AFPMSGs.

3 | MACHINE PERFORMANCE CALCULATION

The back EMF of the coreless AFPMSG at no-load voltage of each phase can be calculated using Faraday's law of electromagnetic induction based on the time variation of the magnetic flux.

$$E_{pb} = \sum_{i=1}^{N_t} \sum_{j=1}^{N_c} \frac{\Delta \phi_j}{\Delta t} \quad (27)$$

$$E_{pb}(\theta_t) = \frac{N_t \sum_{i=0}^{N_c-1} \left(\int_{R_{mi}-iD_c}^{R_{mo}+iD_c} \int_{\theta_t-i\theta_c}^{\theta_t+\theta_{in}+i\theta_c} B(r, \alpha) d\alpha dr - \int_{R_{mi}-iD_c}^{R_{mo}+iD_c} \int_{\theta_t-i\theta_c+\Delta\theta}^{\theta_t+\theta_{in}+i\theta_c+\Delta\theta} B(r, \alpha) d\alpha dr \right)}{2\Delta t} \quad (28)$$

where N_t is the number of columns of a single coil in the axial direction and N_c is the number of rows of a single coil. The magnetic flux passes through the coil vertically, which is calculated by the area integral of the magnetic flux density B on the cross-section surface of each coil. The intersection surface and vertical view of a single coil are shown in Figure 8.

Since the electric degree of each turn is different, the total no-load induced voltage of one coil in each phase

needs to be calculated by adding up the electromotive force of each turn of the coil. The expression of no-load voltage of one coil of a single phase can be written as shown in (28). Where θ_i is the electrical degrees that the inner coil spans. And the electrical degree of a single coil span θ_c is given

$$\theta_c = \frac{\pi^2(R_{mo} + R_{mi})}{pD_c} \quad (29)$$

The torque is calculated by

$$T_e = \sum_{N_t N_c} B_g(i) I(t) l_a \quad (30)$$

where $I(t)$ is the current in the coil and l_a is the radial length of each coil. The flow chart of the proposed MEC method is shown in Figure 9

4 | FINITE ELEMENT ANALYSIS

The FEM model of a coreless-stator AFPMSG is built as shown in Figure 10 (a) and the mesh size of different parts of the model is shown in Figure 10 (b). Simulations of the magnetic field distribution and the no-load phase voltage are obtained by using the FEM software Ansoft. The results of the proposed 3-D MEC model are calculated by MATLAB.

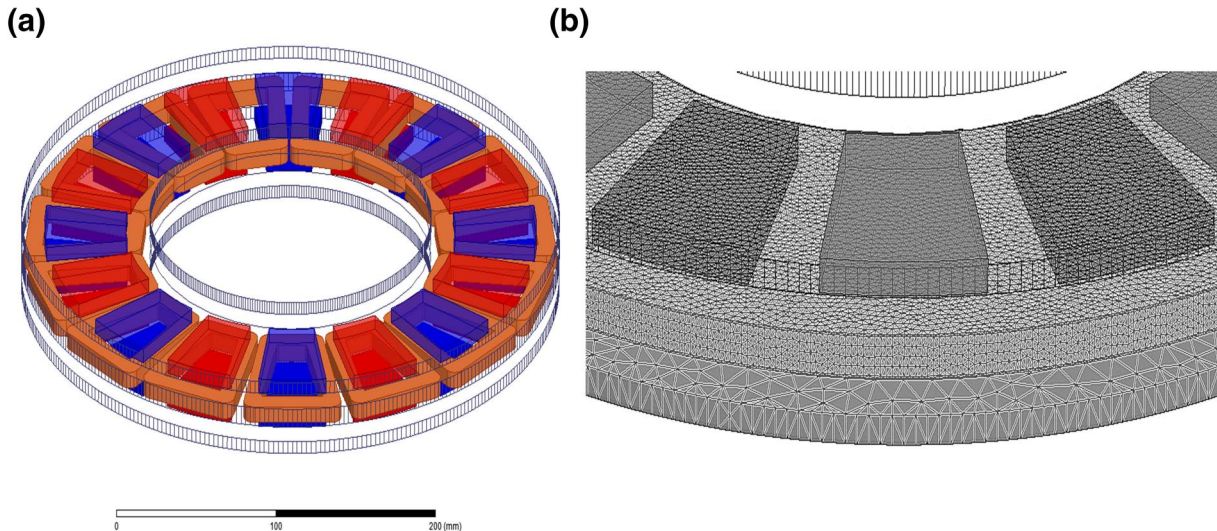


FIGURE 10 Finite elements model of the prototype machine

The left images in Figure 11 show the result of the vertical view of the axial component of the magnetic flux distribution calculated via the MEC model ($N = 50$) with three different axial positions from the mid-air gap to the

surface of PM. The calculation results are plotted using the Matlab software. The simulation results are shown in the right images in Figure 11. The difference between the MEC results and FEM results is mainly at the corner of the flux

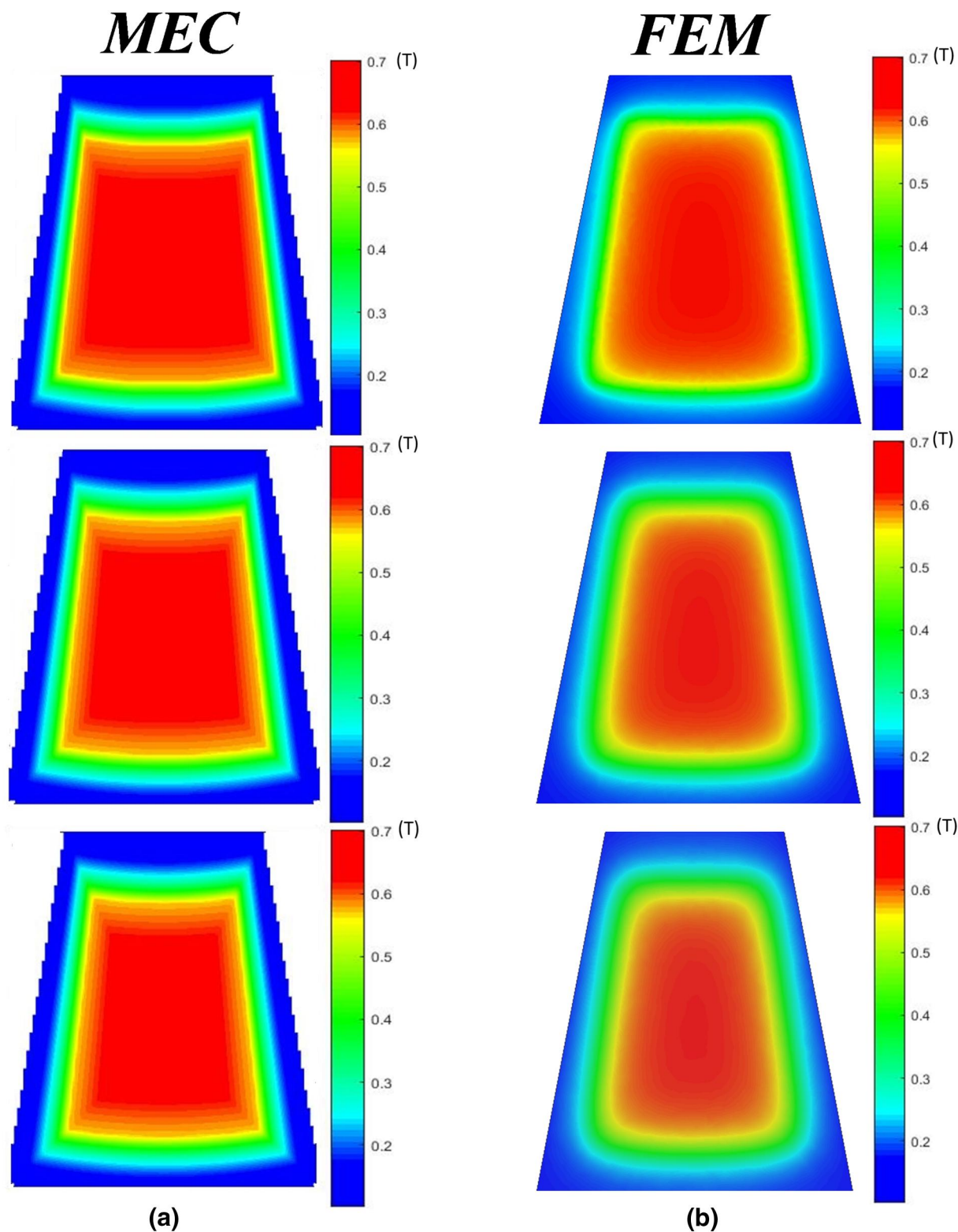
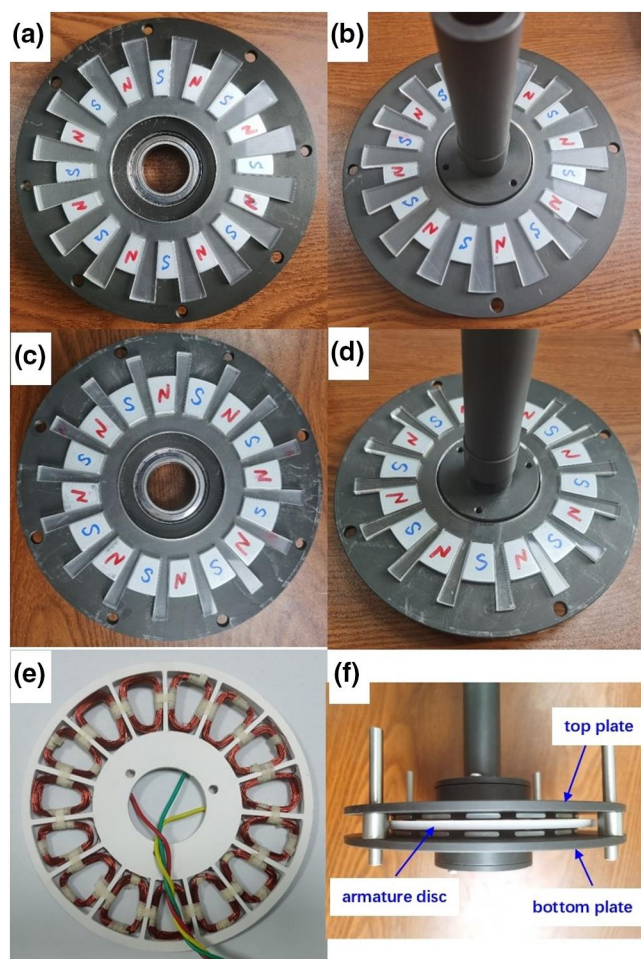


FIGURE 11 Axial component of the magnetic flux distribution in the air gap ($\alpha_p = 0.7$). (a) magnetic equivalent circuit model ($N = 50$), (b) finite elements

TABLE 3 Computational time of the magnetic field calculation for two AFPMSGs

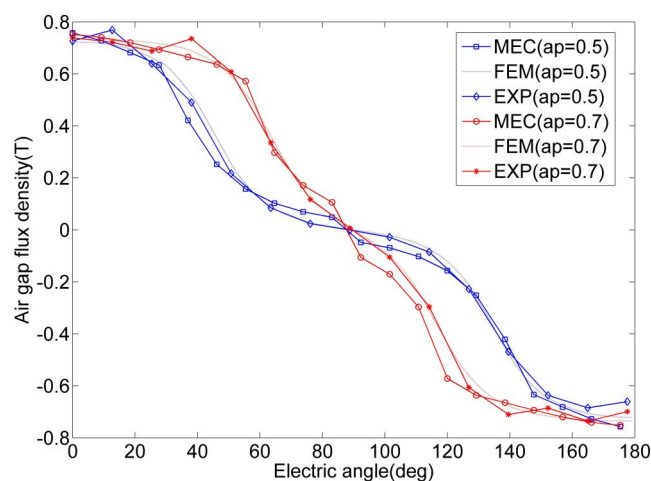
Model	Computational Time (s)	
	3-D FEM	MEC
AFPMSG ($\alpha_p = 0.5$)	15,646	2.3
AFPMSG ($\alpha_p = 0.7$)	17,365	3.1
AFPMSG ($\alpha_p = 0.7$)	17,365	3.1

Abbreviations: AFPMSGs, axial flux permanent magnet synchronous generators; FEM, finite elements; MEC, magnetic equivalent circuit.

**FIGURE 12** Axial flux permanent magnet synchronous generator (AFPMSG) prototype machines with different pole-arc ratios. (a) and (b) top and bottom plants for AFPMSG with $\alpha_p = 0.5$, (c) and (d) top and bottom plants for AFPMSG with $\alpha_p = 0.7$, (e) armature disc, (f) assembled AFPMSG prototype machine

edges, which result from the estimation error of each permeance element shape.

The computational time of magnetic field calculation for two AFPMSGs with different pole-arc ratio ($\alpha_p = 0.5$, $\alpha_p = 0.7$) is listed in Table 3. For two AFPMSG models, the computational time of the proposed MEC model is significantly reduced compared to the computational time of the 3-D FEM method.

**FIGURE 13** Magnetic field measurement system for the axial flux permanent magnet synchronous generator prototype machine**FIGURE 14** The axial component of the magnetic flux density obtained by the magnetic equivalent circuit model, finite elements, and experiments ($N = 10$)

5 | PROTOTYPE VALIDATION

Two AFPMSGs with different pole-arc ratios ($\alpha_p = 0.5$, $\alpha_p = 0.7$) are prototyped for the experimental validation. The parameters in Table 1 are used for AFPMSG prototype manufacturing. The stator and rotor of prototype machines are shown in Figure 12. The rotor includes a top plate and a

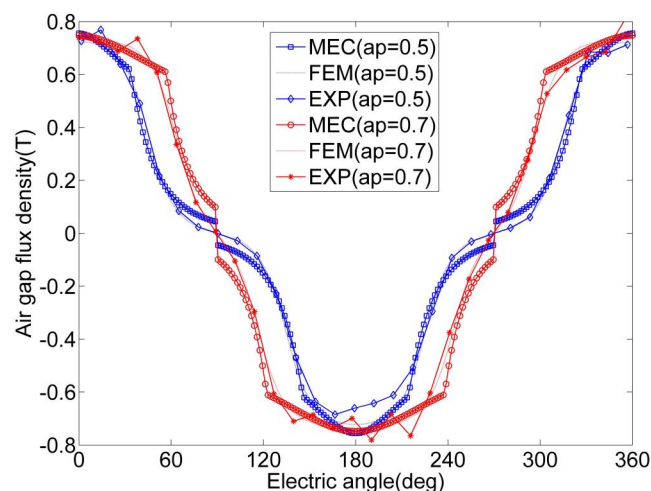


FIGURE 15 The axial component of the magnetic flux density obtained by the magnetic equivalent circuit model, finite elements, and experiments ($N = 50$)

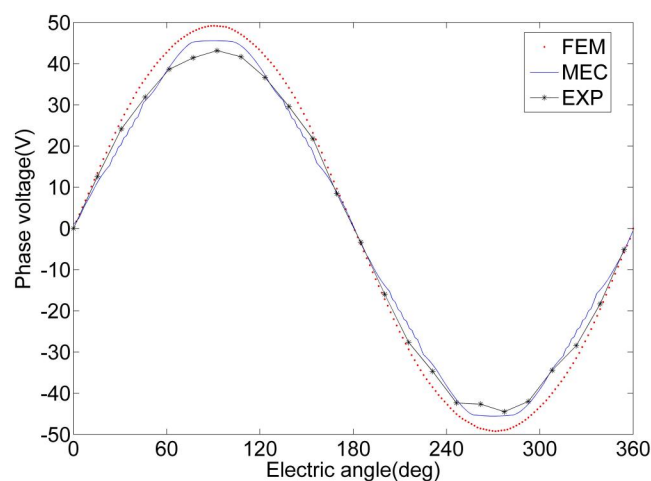


FIGURE 16 No-load back electromotive force at 200 rpm obtained by the 3-D magnetic equivalent circuit, finite elements, and experiments

bottom plate with permanent magnets. In order to measure the magnetic flux density, the prototype machine is not sealed in a cylindrical edge. The magnetic flux density is measured by a magnetic field measurement system in which a gaussmeter is fixed on 3-D movement stage and can move in three dimensions, as shown in Figure 13.

The results obtained by the proposed MEC model are compared with the FEM results and the measured results from the prototypes. The distribution of the axial component of the magnetic flux density is shown in Figure 14 and Figure 15 for two AFPMSGs with different pole-arc ratios. The results show a good agreement between the MEC, FEM, and experimental results of the magnetic field distribution. The back EMF waveform under no-load phase voltage condition of at 200 rpm is measured and shown in Figure 16. The effective value of back-EMF is shown in Figure 17. The calculated values of the back-EMF are slightly smaller than the results of

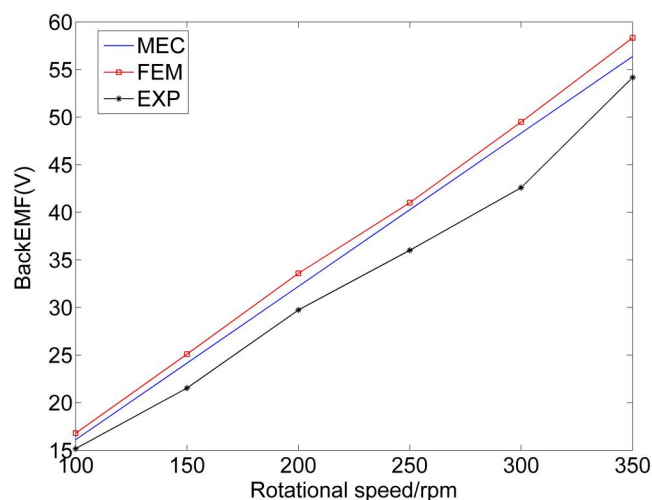


FIGURE 17 The effective value of no-load back-electromotive force at different rotation speeds obtained by the 3-D magnetic equivalent circuit, finite elements, and experiments

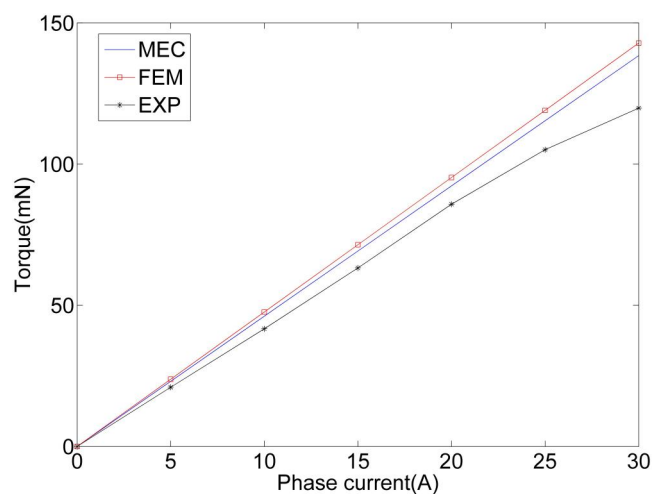


FIGURE 18 The magnitude of torque at different phase current obtained by the 3-D magnetic equivalent circuit, finite elements, and experiments

the MEC and FEM. The reason may be that the shape of each coil in the armature disc is slightly different from that used in the MEC and FEM.

The axial component of the magnetic flux density at the centre of the air gap with two pole-arc ratios ($\alpha_p = 0.5$, $\alpha_p = 0.7$) is shown in Figures 14 and 15. It can be seen that the axial component magnetic flux density distribution obtained by the proposed 3-D MEC model has good agreement with the FEM and measured results.

When $N = 10$, the maximum errors between the calculated and measured magnetic flux density are 8.1% and 7.4% for $\alpha_p = 0.5$ and $\alpha_p = 0.7$, respectively. When $N = 50$, the maximum errors between the calculated and measured magnetic flux density are 7.8% and 6.6% for $\alpha_p = 0.5$ and $\alpha_p = 0.7$, respectively. It indicates that the accuracy of the proposed 3-D MEC is enhanced by increasing N . The limitation of the

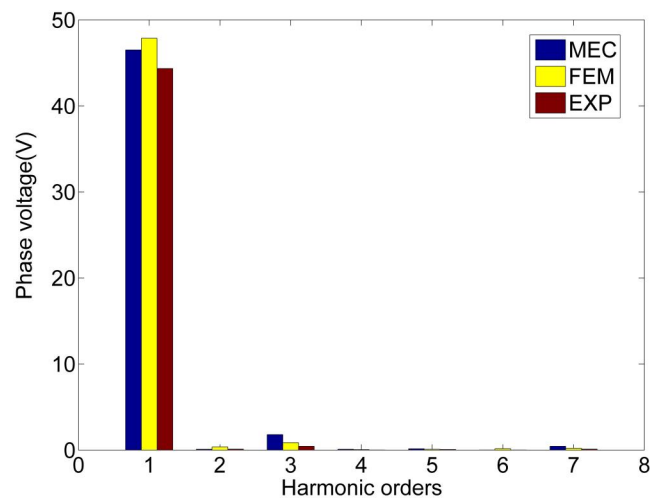


FIGURE 19 Harmonic components of no-load phase voltage obtained by 3-D magnetic equivalent circuit, finite elements and experiments

proposed method leads to a sudden change of flux density at the centre of adjacent poles (electric angle close to 90). It indicates that it is possible to obtain a relatively accurate magnetic field distribution in the large air gap by using the proposed 3-D MEC method.

The no-load back-EMF waveform at 200 rpm is shown in Figure 16. It can be noticed that the MEC model gives lower no-load back-EMF than the FEM results, but gives a higher no-load back-EMF than the experimental results. The maximum error between the calculated and measured no-load back-EMF is less than 7.6%.

The effective values of no-load back-EMF at different rotation speeds obtained by the 3-D MEC, FEM, and experiments are shown in Figure 17. It can be seen that no-load back-EMF increases almost linearly with increasing rotation speed. The MEC results show good agreement with the FEM and experimental results. The values of no-load back-EMF obtained by the MEC model are slightly lower than the FEM results and higher than the experimental results. The maximum error between the calculated and measured no-load back-EMF is 11.3% at 300 rpm.

The torques at different phase current obtained by the 3-D MEC, FEM, and experiments are shown in Figure 18. The magnitude of the harmonic components of no-load phase voltage is shown in Figure 19. The good agreement between the calculated and measured results validates the effectiveness and good accuracy of the proposed 3-D MEC model.

6 | CONCLUSION

In this paper, a 3-D MEC model for the magnetic field calculation in coreless AFPMSGs is proposed by considering the magnetic leakage and fringing flux. The magnetic flux in the air gap between the permanent magnets is analysed using

the magnetic field division method. The magnetic flux between the PM poles in the air gap is divided into onion-shaped magnetic flux tubes. The magnetic permeance of the main magnetic flux is calculated based on the proposed 3D MEC model. The magnetic leakage from the PM to the rotor iron plates is considered in both radial and tangential directions. The proposed MEC method is applied to a specific AFPMSG, and the magnetic field distribution, no-load back-EMF, and torque are calculated. The 3D FEM model of the AFPMSG machines is prepared to confirm the validity of the proposed MEC model. AFPMSG machines with different pole-arc ratios ($\alpha_p = 0.5$ and $\alpha_p = 0.7$) are prototyped to validate the effectiveness and accuracy of the proposed 3-D MEC model. The results obtained by the proposed MEC model are in good agreement with those of the FEM and experimental results.

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